# Sep 8 warmup

Find $\int \sec(x)\,dx$

$\sec = \frac{1}{\cos}$

$\sec x = \frac{\sec(x)\,(\sec x + \tan x)}{\sec(x) + \tan(x)}$

$\int \frac{\sec(x)(\sec x + \tan x)}{\sec x + \tan(x)}\,dx$

$= \int \frac{\sec^2(x) + (\tan x \sec x)}{\sec x + \tan x}\,dx$

$u = \sec x + \tan x$

$du = \sec x \cdot \tan x + \sec^2 x\,dx$

use u-sub

$\int \frac{\sec^2(x) + (\tan x \sec x)}{\sec x + \tan x}\,dx = \int \frac{1}{\sec x + \tan x} \cdot (\sec x \cdot \tan x + \sec^2(x))dx$

$= \int \frac{1}{u}\,du = \ln|u| + C = \ln|\sec x + \tan x| + C$

$= \ln|\sec x + \tan x| + C$

Trig identities

$\frac{d}{dx}(\sin x) = \cos x \qquad \frac{d}{dx}(\cos x) = -\sin x$

$\frac{d}{dx}\tan x = \sec^2 x \qquad \frac{d}{dx}\sec x = \sec x \tan x$

$\sin^2(x) = \frac{1}{2}(1 - \cos 2x)$

$\cos^2(x) = \frac{1}{2}(1 + \cos 2x)$

$\sin^2(x) + \cos^2(x) = 1$

$\tan^2(x) + 1 = \sec^2(x)$

Corollary: $1 = \sec x + \tan x$